Smart Thoughts: BCI Based System Implementation to Detect Motor Imagery Movements

Ikram Mustafa, Imran Mustafa
CESAT, Islamabad
ikrammustafaa@yahoo.com,imranmustafa90@yahoo.com

***Abstract*— Brain-Computer Interface (BCI) is an approach for connecting human brain and a computer or any other type of electronic devices such as the robot and automated wheelchair, etc. The main purpose of BCI is not to trust only on peripheral nerves and muscles for communication because sometimes they are not in the condition to perform tasks or giving signals to devices, so that's why BCI is directly connected to the brain for activity recording. The BCI based system needs to record brain signals to perform operations, these signals can be recorded with Electroencephalogram (EEG) technology. EEG signals are recorded neural activity of the brain, which occurs when a user is performing some type of task like focusing on something or trying to give motor imagery signals to muscles. The purpose of this research is to enable the user to send motor imagery movements commands to the computer and make sure that the computer will perform these motor imagery movements correctly and consume less time for real-time processing. For ensuring all these things an algorithm (not only algorithm a way for BCI systems to record, filter and process signals smartly) is designed. In the proposed approach, major electrodes dealing with motor imagery movement are selected and others are eliminated, which make proposed system more effective light and fast after eliminating extra electrode signals are passed by the band-pass filter for eliminating contaminated signals. Results show that proposed system is increasing the accuracy of motor imagery movement detection. Eliminating the extra data efficiently, which is making the system more light and fast. By making the number of electrodes less proposed system is optimizing the signals which enabling real-time processing much better, fast and accurate. The main purpose of the proposed approach is to make BCI based motor imagery movement detection, better, fast, efficient and performing real-time operations. Proposed work has many avenues for the future.**

Keywords — Brain-Computer Interfaces (BCI), Electroencephalography (EEG), Motor Imagery movement, Emotive Epoc+, SVM in BCI system.

I. INTRODUCTION

Computers have traced nearly every aspect of our lives nowadays, from performing the critical function in mathematics or solving major problems in the industry, handling billions of records, training, entertainment, medicines, surgeries, and work. In last few years Brain-Computer Interfacing (BCI) research came up with the new level of understanding and relatively low cost and powerful equipment and majorly recognition of the needs of disable and paralyzed persons. BCI allows brain-derived communication in paralyzed patients or locked in patients all we required is some deliberate muscle control. For numerous inordinate lengths of time, the researchers are trying to make communication between human brain and external world by using electroencephalographic measures of brain functions.

Brain-Computer Interface is a mode of connecting human brain and a computer or any other type of electronic devices such as the robot and automated wheelchair, etc. The main purpose of BCI is not to trust only on peripheral nerves and muscles for communication because sometimes they are not in the condition to perform tasks or giving signals to devices, so that's why BCI is directly connected to the brain for activity recording [1]. There are two major requirements which are compulsory to communicate between a computer and human brain is:

1) Structures that are valuable to differentiate between several kinds of human brain states [1].
2) Methods and algorithms for detection and sorting these states of the human brain in real time environment [1].

In this paper, research is primarily on the BCI based system, for that some strong and properly structured definition of the brain-computer interface will be very useful. Following is the quote from the first international BCI conference [2]

"A brain-computer interface is a communication system that does not depend on the brain's normal output pathways of peripheral nerves and muscles." [2]

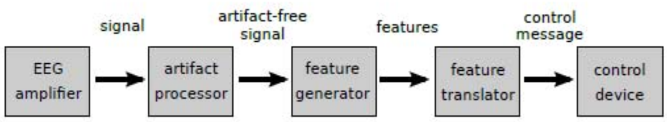

Figure 1 Classic Model of Available BCI System [2]

A classic model of BCI is clearly defined in above figure which comprises of following elements

- **EEG based set of Electrode and the Amplifiers:** These electrodes and amplifiers are responsible for the detection of brains neural electrical activity in form of signals [2].

978-1-5386-3564-3/18/$31.00 ©2018 IEEE

- **Processor Used for removing Artifacts:** In between the Recording of EEG signals, we get a lot of noises and artifacts which can be generated by other brain activates happening in the background or utility frequency artifacts. This processor is responsible for the reduction of these artifacts and noises from the generated signals [2].
- **Feature Generator:** these generators are mainly responsible for the transformation of the raw signals of EEG into a set of the stable and understandable features which can be used by classifiers for feature extraction and classification [2]. This module can be divided into sub-modules for execution of the behind tasks
  - Pre-Processing: In this phase, raw signals from EEG is processed with the help of processing algorithms in order to improve the signal to noise ratio [2].
  - Feature Extraction: the major purpose of feature generator is feature extraction, this part contains algorithm's which are used for extracting the major signal of features from the raw signals [2].
- **Feature Translator:** feature translator mainly contains feature classification and post-processing and used to decode the signals which are acquired in the earlier step of feature generation these signals are sent to control device for task performance [2].
- **Control Devices:** these are the devices which are used to map the signals acquired and filtered by EEG [2].

II. Literature Review

In this section of the paper, existing techniques for the BCI Is discussed. These existing techniques are discussed to establish a background context for the developed technique in the presented technique. By describing the current literature, this section of paper intends to highlight limitations of the existing approaches. Existing techniques for classification and feature extraction is also discussed and select classification and feature extraction techniques for the proposed approach.

Electroencephalography (EEG) is a method used for the recordings of human brains electrical neural activity of human brain with the help of electrodes which are placed on the top of the scalp in the form of 10/20 system. EEG is a non-invasive technique of mapping signals from the human mind, a direct amount of cortical movement with a temporal resolution of smaller than a millisecond is only provided by EEG. EEG is the method which can also be used to collect the structures of brain signals even if the person is not in the condition to attend to stimuli. Functional Magnetic Resonance Imaging, Magneto Encephalography (MEG), Near-infrared Spectroscopy (NIRS), Positron Emission Tomography (PET) and Event-related optical signal (EROS) are some famous Methods to monitor human brain function [3].

Existing Pre-processing and Classification Techniques for BCI:

P300, **Steady-State Visually-Evoked Potentials (SSVEP)** is, **Slow-cortical potentials (SCP)**, **Conversion of EEG Activity into Cursor Movement**, **Event-Related Desynchronization (ERD)** is another BCI technique for pre-processing and classification technique for EEG based signals. In ERD changes occurring in the power of signals of mu bands (8 to 13 Hz) and beta bands (13 to 30 Hz) are detected and motor-imagery movement detection is predicted with the help of an algorithm. ERD method is being used in proposed approach[4].

Existing Classifiers for Feature Extraction Techniques for BCI

Random forest[5], **Bayesian Classifier[5], Neuro-Fuzzy[5]**, **Time-domain parameters[5], Linear Discriminant[5], Support vector machines SVM[5]** is a widely used kernel-based method for feature extraction in motor imagery based BCI systems. SVM is developed in three stages 1) SVM is extended to hyperplane into a feature space prompted by kernel-function 2) SVM covers nonlinear restrictions between classes 3) SVM prepared to address noisy-data (artifacts) which make signals distorted or violating margins between 2 classes [6].

Hyperplanes for SVM is calculated with the help of the following equation.

$y = wT\ x + b$ ……………….. (2)

And kernel function of SVM is calculated with

$K(x,y)= e\ \_\ ||\infty\text{-}x||2/(2.6)2$ ……………….. (3)

III. Data Set

A. Description of Data Set

Experiments performed in this paper is based on BCI competition's datasets. BCI competition is an open competition for BCI based researchers whose goals are evaluating numerous methods which are currently being used in the field of Brain-Computer Interfacing, and after comparing them deriving the best method among all the methods for evaluating and obtaining best and reliable dimensions of the all algorithms on the basis of performance and accuracy for BCI related signal processing [7].

BCI competition has four editions till date (the latest one is in the year 2008). Each edition of BCI competition is comprised of 3 to 5 data sets all the data sets are freely available for researchers. Each data set is distributed into two portions one is training data set and other is test data set. The goal is to provide programs or code or output methods to find best and as accurate as possible results or signals that could be capable to appropriately categorize the test data set. In our research, we are using BCI competition IV data set 2a in .gdf format.

Our dataset contains EEG based signal recordings from 9 subjects. Every subject has 2 sessions recording on different days. Every session comprised of 6 rounds each round consists

of 48 streams 12 streams for every motor imagery class. In every round, a cue was displayed on screen teaching the subject to complete specific motor imagery task: left-hand, right-hand, both-feet or tongue-movement.
Every single round comprised of 288 trails in total, 72 for every single motor imagery movement based task. Each trail is started with a short warning sound and at the same time when warning sound is being played a fixation-cross appears on screen, after 2 seconds on screen at the place of fixation-cross a cue will be displayed which shows the arrow, according to that arrow subject has to start the corresponding motor imagery movement. After1:25 seconds the arrow will be converted back into fixation-cross again and the subject has to continue the motor imagery movement for six seconds and then after six seconds fixation-cross will disappear and there will be a short break of 1:5 seconds.

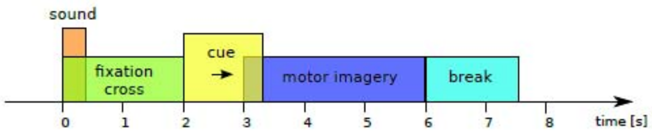

Figure 2 Timeline of signal trails from BCI competition data set

The data of data set was logged by using 22 EEG-Electrodes and 3 EOG electrode (EOG electrodes are used for measuring eye-bases movement detection). Signals were recorded at 250Hz sampling-frequency and 0:1 to 100Hz band pass filtered and 50Hz notch-filtered. Placement of EEG based electrodes are based on 10-20 system which is consisted of Fz, C3, C4, Cz and Pz electrodes and other electrodes are placed at 10% distance away from the stated electrodes.[8].

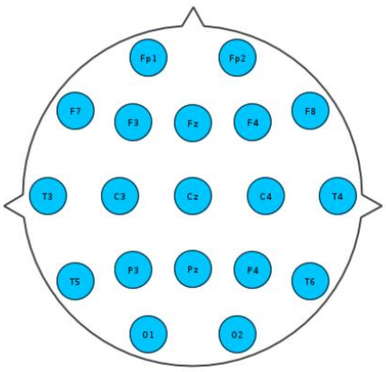

Figure 3 10/20 System's Electrodes placements

B. Preprocessing

Pre-processing used for dataset involves the following steps:

1. Re-referencing of the recorded EEG signals
2. Bandpass purification of EEG signals
3. Re-Sampling of the EEG signals
4. Separations of EEG signals
5. Choosing the clean EEG signals

Common methods for referencing signals are laplacian-filtering [9]. The purpose is to find a neutral place and reducing the impact of the noise and artifacts. Application of these techniques is very helpful for finding good quality of the gathered EEG signals [9-10].
EEG signals recorded from the brain are bandpass before the analysis of the data acquired. This bandpass is depending on the range of frequencies of interest and the projections of signal analysis. Filter features can be different. They can possibly use high pass filtration range (for e.g., 0.1 Hz) for event-related potential (ERP's). These frequency bands are used in between spectral analysis (e.g., 8-13 Hz for the monitoring of the activity of alpha signals) [11]. A power-line-notch filter is applied for removing artifacts slowing down the electrical instruments in the locality. It should be noticed here that improper filtration implementation can harm EEG recordings.
After applying proper filter we re-sample our EEG data, re-sampling data is normally an offline process. Samples are in the range of 4 kHz but only EEG signals which are less the 100Hz has been selected. Usage of re-sampling can reduce the noises and very much useful for saving the transmission power in the case of real-time designing of the system.
After successfully applying all above-mentioned steps the major step that is finding the clean EEG signals or segments from our EEG recordings has been started. For extraction of clean EEG signals we can use two methods: 1) Clean the segments manually by visualizing of signals (visualizations can be performed by the clinical experts who knows about the signals) 2). Clean the segments automatically using signal processing method. A hybrid system may be used that is the combination of both manually or automatically. In the proposed approach, a hybrid method has been used. After inspection of the signal, we remove contaminated signals and select clean signals only.

C. Noise (Artifacts) Removal

Noise or artifact is terms used for improper data signals in EEG recordings, which can be generated by internal brain activities happening in the background of the brain or because of external activities like environmental disturbance or signals generated by EEG equipment etc.
These noises or artifacts are removed with the help of filtration methods available in the toolbox (Biosig toolkit for MATLAB) which is being used in research.

D. Classification of Data

Classification of the data is performed by unsystematically selection of the 4 random trails for each movement which is used as the test data and the 14 left behind trails are used as training data. In maximum cases, diagonal-linear and diagonal-quadratic representations of classifiers are used and finest results were achieved from them. Classification procedure is executed thousands of times for each and every motor imagery movement then the average of all trails is calculated for the final results.

E. Feature Extraction

After applying classification, we have specific motor imagery movement on the specific electrode which can be used for movement detection.
Movement detection according to electrode is defined below

C3 electrode: Right motor imagery Movement
C4 electrode: Left motor imagery Movement
Cz electrode: Foot motor imager Movement
Fz electrode: Foot motor imagery Movement
Pz electrode: Tongue motor imagery Movement

Our data set have following events for each electrode where motor imagery movement is being detected.

Event	type	Description
276	0x0114	Idling EEG (eyes open)
277	0x0115	Idling EEG (eyes closed)
768	0x0300	Start of a trial
783	0x030F	Cue unknown
1023	0x03FF	Rejected trial
32766	0x7FFE	Start of a new run

IV. Evaluation Method

Dataset 2a of BCI competition IV is calculated and matched by using kappa-coefficient in Matlab by using BioSig toolbox.
Kappa-coefficient a takes 0 as input for a random-classifier and 1 for the perfect-classifier that will always classify correctly.
Value for the kappa-coefficient is calculated by using the following equation

$$K = \frac{p0 - pe}{1 - pe}$$

Where, p0 is classification-accuracy and pe is hypothetical-accuracy of the random-classifier of the similar data.
In case of our data set pe=0.25

$$K = \frac{p0 - 0.25}{1 - 0.25}$$

$$K = \frac{p0 - 0.25}{0.75}$$

Results are generated with the help of above equation are calculated for each trail in the sample. After that, confusion matrices and the kappa-value are constructed individually for each trail sample. The final performance measure is the maximum value of kappa-value from the considered time course. Implementation of the code of the kappa-algorithm is available in BioSig Libraries documentation [12-13].

V. Results

Dataset 2a of BCI competition 4 is comprised of 18 files in total, each file is holding a recording of single-trial session for a particular subject, and one session for a particular subject is labeled for an individual trial. With the help of this, we know that which motor imagery based task is being performed by the subject and that session is projected to be used as training set.
And for the all others sessions no label is provided which is being used as test data. We have to label it correctly after identifying the correct motor imagery based movement and that data set contains all the artifacts too which also is to be removed for obtaining desired results.
The output should classify the correct motor imagery movement by using asynchronous BCI based system and classify the EEG signals into the correct intentions of the subject at the particular time.
EEG signals data set is acquired with the help of twenty-two Ag/AgCl electrodes having the distance of 3.5cm of inner electrodes. All acquired signals were recorded and documented monopolarly with the help of left-mastoid which is working as a reference and the right-mastoid which is working as ground.
All signal samples were taken with the frequency of 250Hz and then the band-passed filter is applied between the frequencies of 0.5Hz to 100Hz. Amplifier's sensitivity is set to 100 µV. An additional 50Hz notch-filter was used to reduce the line-noise.
3 mono-polar EOG electrodes are also recorded and sampled at 250Hz to help 22 EEG channels. These EOG electrodes data is band-pass filtered between the frequencies of 0.5Hz to 100Hz and Amplifier's sensitivity is set to 1 µV. The EOG channels are delivered for the consequent application of artifact processing procedures [1 desc 2a paper] and strictly not be used for the purpose of classification.

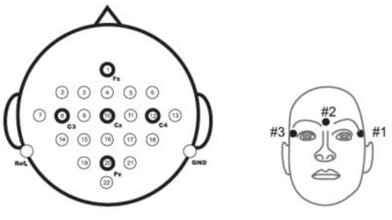

Figure 4 Left: Electrodes placements of worldwide 10-20 System. Right: Electrodes placements of three mono-polar EOG channels.

The classifier of data which is being used for the classification purpose is Support vector machines (SVM). SVM is a function which is used for dividing space (in this case it is data set) into two major parts classifications class and results. In this paper with the help of SVM, function data is being divided into two parts classification with events and their results.

In dataset for classification, we are focusing on specific electrodes for classifying data for our motor imagery movement in our data set we are focusing on Fz, Cz, Pz, C3, and C4.

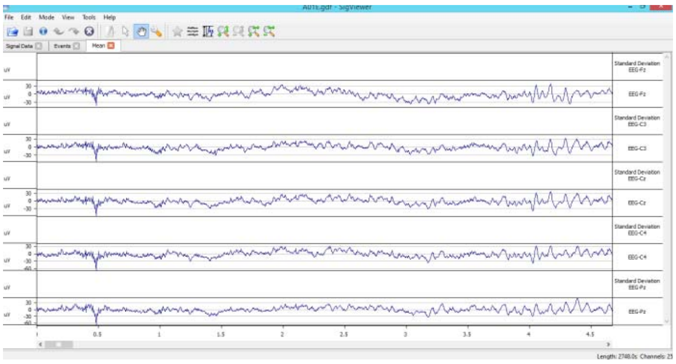

Figure 5 Classification of data

In the figure, the specific electrodes are filtered for the detection of our major motor imagery movement signals.

Right motor Imagery Movement

For first motor imagery movement, we have c3 electrodes signals which are showing our right motor imagery movement.

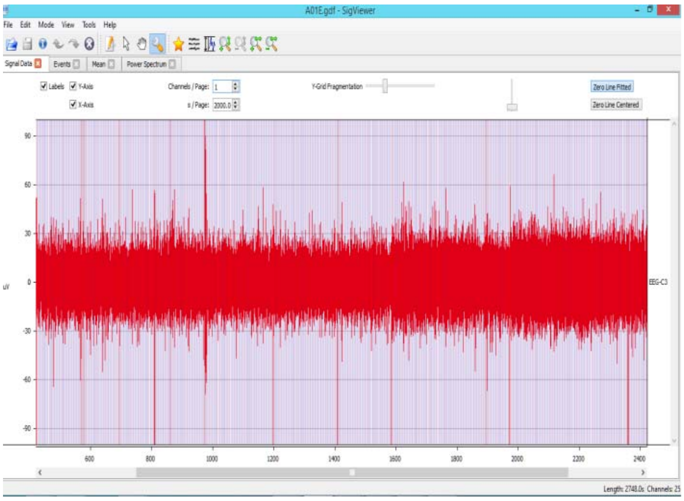

Figure 6 Average of the signals of Right imagery movement

Above image is showing the average of the signals of right motor imagery movement and the frequency value of the signal is between -30Hz to 30Hz signals above and below this frequency is neglected because of the average.

Left motor Imagery Movement

For 2nd motor imagery movement, we have c4 electrodes signals which are showing our left motor imagery movements signals.

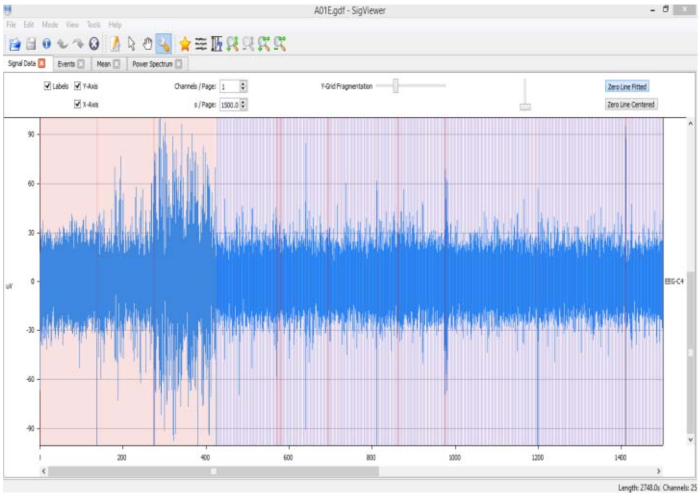

Figure 7 Average of the signals of left imagery movement

Above image is showing the average of the signals of left motor imagery movement and the frequency value of the signal is between -60Hz to 60Hz signals above and below this frequency is neglected because of the average.

Foot motor Imagery Movement

For 3rd motor imagery movement, we have Cz and Fz electrodes signals which are showing our foot motor imagery movements signals. Signals from the foot are considered as signal movement detection because foot signals are very much same and very hard to classify differently because of the same origin of signal generation.

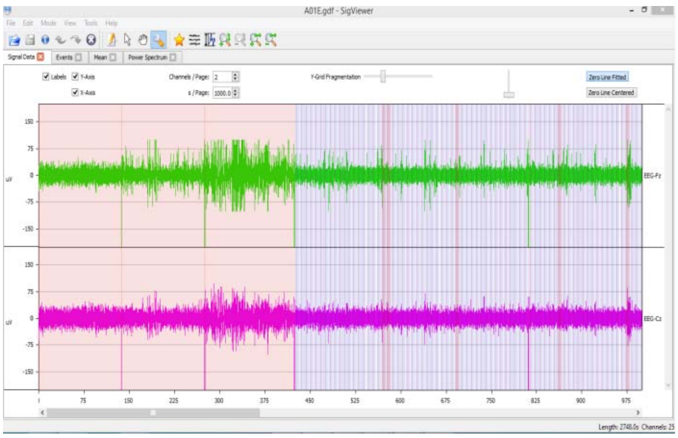

Figure 8 Average of the signals of foot imagery movement

Above image is showing the average of the signals of foot motor imagery movement and the frequency value of the signal is between -75Hz to 75Hz signals above and below this frequency is neglected because of the average.

Tongue motor Imagery Movement

For 4th motor imagery movement, we have Pz electrode signals which are showing our tongue motor imagery movements signals.

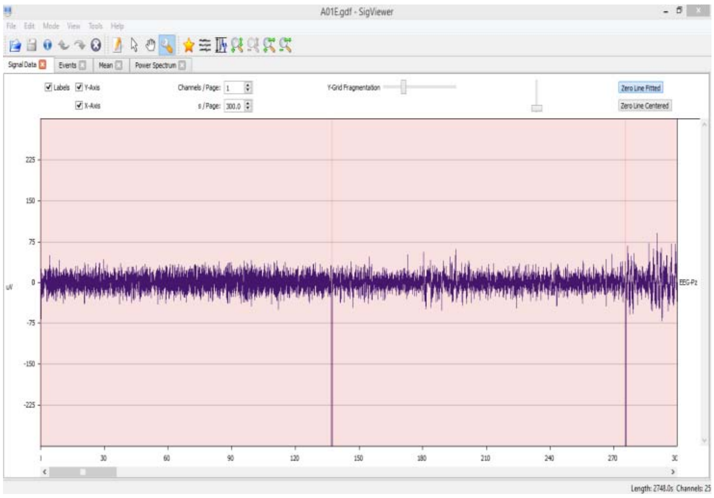

Figure 9 Average of the signals of Tongue imagery movement

Above image is showing the average of the signals of left motor imagery movement and the frequency value of the signal is between -60Hz to 60Hz signals above and below this frequency is neglected because of the average.

Overall Results

All above results were obtained by using SVM and cross-validation technique on our BCI IV competition dataset.
By using following Band-pass parameters results are generated.

- Trails trimmed to 3:55 sec to 6:00 sec,
- Size of smoothing windows is 2 seconds,
- Frequency-bands used are 8 to 14 Hz, 19 to 24 Hz and 24 to 30 Hz.

Time-Domain Based Parameters

- Trails trimmed to 4:01 sec to 6:00 sec,
- Five derivates are used,
- Average exponential-windows size is 0:0155.

Entire data was down-sampled by the factor at 4 and filtered by 7 to 30 Hz.
SVM-classifier used C:б pairs
C = 1, б = 100 when used with band-power based features,
C = 100, б = 100 when used with time-domain based parameters.

Features	Classifiers	Kappa values for all Subjects									Sum of All Subjects	Mean Kappa Value
		1	2	3	4	5	6	7	8	9		
C3: Right	SVM	0.86	0.88	0.83	0.86	0.87	0.81	0.83	0.82	0.88	7.64	0.848889
C4: Left	SVM	0.85	0.87	0.80	0.83	0.85	0.89	0.81	0.89	0.82	7.61	0.845556
CZ: foot	SVM	0.84	0.84	0.82	0.80	0.83	0.82	0.81	0.85	0.80	7.41	0.823333
Fz: foot	SVM	0.85	0.87	0.80	0.80	0.81	0.80	0.89	0.83	0.89	7.54	0.837778
Pz : Tongue	SVM	0.84	0.86	0.89	0.87	0.89	0.84	0.82	0.87	0.81	7.69	0.854444

Table 1 Final Kappa value for all subjects using SVM Classifier

In table 1 Kappa values by using SVM classifier are shown. First calculated Kappa value for each subject by applying following steps:
First, Band-pass parameters results are generated. Trails trimmed to 3:55 Sec to 6:00 Sec, Size of smoothing windows is 2 seconds, Frequency-bands used are 8 to 14 Hz, 19 to 24 Hz and 24 to 30 Hz.
Than Time-Domain Based Parameters calculated, Trails trimmed to 4:01 Sec to 6:00 Sec, Fivederivate's are used, Average exponential-windows size is 0:0155.Entire data was down-sampled by the factor in 4 and filtered from 7 to 30 Hz.
SVM-classifier used C: б pairs
C = 1, б = 100 when used with a band-power based features,
C = 100, б = 100 when used with time-domain based parameters.
After the SVM classifier on each individual electrode, then the sum and mean of kappa values are taken for averaging the kappa value of electrode for accurate measurements.

Features	Classifiers	Kappa values for all Subjects									Sum of All Subjects	Mean Kappa Value
		1	2	3	4	5	6	7	8	9		
C3 : Right	LDA	0.84	0.75	0.82	0.70	0.75	0.77	0.81	0.81	0.75	7	0.777778
C4: Left	LDA	0.82	0.75	0.87	0.71	0.72	0.86	0.77	0.75	0.70	6.95	0.772222
Cz : foot	LDA	0.81	0.70	0.80	0.76	0.70	0.80	0.79	0.71	0.78	6.85	0.761111
Fz : foot	LDA	0.81	0.63	0.79	0.77	0.76	0.77	0.75	0.70	0.75	6.73	0.747778
Pz : Tongue	LDA	0.81	0.63	0.81	0.84	0.75	0.84	0.71	0.75	0.77	6.91	0.767778

Table 2 Final Kappa value for all subjects using LDA Classifier

In table 2 Kappa values by using LDA classifier are shown. First calculated Kappa value for each subject by using LDA classifier on each individual electrode, then sum and mean of kappa values are taken for averaging the kappa value of electrode for accurate measurements.

Features	Mean Kappa values for all Subjects			
	Mean Values LDA	Mean Values SVM	LDA Accuracy in %	SVM Accuracy in %
C3 : Right	0.777778	0.848889	77.7 %	84.4 %
C4: Left	0.772222	0.845556	77.2 %	84.5 %
Cz : foot	0.761111	0.823333	76.2 %	82.3 %
Fz : foot	0.747778	0.837778	74.0 %	83.7 %
Pz : Tongue	0.767778	0.854444	76.5 %	85.6 %

Table 3 Comparison of LDA and SVM Classifiers

In table 3 means the value of each electrode by using LDA and SVM classifier is shown, then accuracy in % is also shown

Figure 10 LDA vs SVM accuracy chart

In table 1 SVM classifiers kappa values, the sum of all subjects' data for specific electrodes and mean values for all subjects' are shown. In table 2 LDA classifiers kappa values, the sum of all
subjects' data for specific electrodes and mean values for all subjects' are shown. In table 3 comparison between SVM and LDA classifier is shown in table 3 difference between the accuracy is also shown. In table 3 accuracy difference between SVM classifier (Used in proposed approach) and LDA classifier can be visibly seen and on the basis of these results figure 10 showing graph with difference of accuracy between LDA and SVM, hence it is proved that proposed approach is more accurate then LDA classifier that is widely used in Motor imagery based BCI system as classifier for feature extraction.

VI. CONCLUSION

The method used in this paper for motor imagery detection is tested on sample data set for better, reliable and fast results. This method allows the person to perform normal motor imagery movements which are performed in daily routine without any special type of training. This approach will detect movement and map that movement to BCI devices for task execution.

This approach is using 10/20 electrodes placement system which allows the person to just focus on some specific electrodes for movement detection, not like traditionally used 64 electrodes method. Detection of motor imagery movement is done only by focusing on 5 electrodes.

Results after performing experiments on dataset show that proposed approach is more effective, reliable and fast for motor imagery movement detection with fewer numbers of electrodes then traditional approach used for motor imagery movement detection.

VII. REFERENCES

[1]. Ming Cheng*, Xiaorong Gao, Shangkai Gao “Design and Implementation of a Brain-ComputerInterface With High Transfer Rates”, IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING, VOL. 49, NO. 10, OCTOBER 2002

[2].A hybrid EEG-fNIRS BCI: Motor imagery for EEG and mental arithmetic for fNIRS. M. Jawad Khan; Keum-Shik Hong; Noman Naseer; M. Raheel Bhutta Control, Automation, and Systems (ICCAS), 2014 14th International Conference on Year: 2014 IEEE Conference Publications

[3].Classification of motor imagery tasks with LS-SVM in EEG-based self-paced BCI Mahmoud E. A. Abdel-Hadi; Reda A. El-Khoribi; M. I. Shoman; M. M. Refaey Digital Information Processing and Communications (ICDIPC), 2015 Fifth International Conference on Year: 2015 IEEE Conference Publications.

[4]Active transfer learning and selective instance transfer with active learning for motor imagery based BCI
Ibrahim Hossain; Abbas Khosravi; Saeid Nahavandhi
2016 International Joint Conference on Neural Networks (IJCNN) Year: 2016 Pages: 4048 - 4055, DOI: 10.1109/IJCNN.2016.7727726 IEEE Conference Publications.

[5].Voting Strategy to Enhance Multimodel EEG
Based Classifier Systems for Motor Imagery BCI Miguel Almonacid; Julio Ibarrola; Jose-Manuel Cano-Izquierdo IEEE Systems Journal Year: 2016, Volume: 10, Issue: 3 Pages: 1082 - 1088, DOI: 10.1109/JSYST.2014.2360433 IEEE Journals & Magazines.

[6].Information based source number estimation for probabilistic common spatial pattern in motor imagery BCI system
Asghar Zarei; Farnaz Ghassemi; Mohammad Hassan Moradi 2016 24th Iranian Conference on Electrical Engineering (ICEE) Year: 2016 Pages: 555 - 560, DOI: 10.1109/IranianCEE.2016.7585584 IEEE Conference Publications

[7]. Jonathan R. Wolpaw, Niels Birbaumer, Wiliam J. Heetderks, Denis J. McFarland, P. H. Peckham, G. Schalk, E. Donchin, L.A. Quatrano, C. J. Robinson, and T. M. Vaughan. "Brain-computer interface technology: a review of the first international meeting". Rehabilitation Engineering, IEEE Transactions on, 8(2):164 –173, Jun 2000.

[8]. Ali Bashashati, Mehrdad Fatourechi, Rabab K. Ward, and Gary E. Birch. "A survey of signal processing algorithms in brain-computer interfaces based on electrical brain signals". Journal of Neural Engineering, 4(2):32–57, 2007.

[9] Paul Sajda, Adam Gerson, K.-R. Muller, Benjamin Blankertz, and Lucas Parra.“A data analysis competition to evaluate machine learning algorithms for use in brain-computer interfaces.” Neural Systems and Rehabilitation Engineering, IEEE Transactions on,11(2):184–185, 2003.

[10]. Clemens Brunner, Robert Leeb, Gernot R. M¨uller-Putz, Alois Schl¨ogl, and Gert Pfurtscheller. BCI Competition 2008 — Graz dataset A. http://www.bbci.de/competition/iv/desc_2a.pdf. Accessed March 1, 2013.

[11]. A. Rousselet "Does filtering preclude Us from studying ERP time-courses?" Frontiers Psychology, vol. 3, no. 131, 2012

[12]. E. Niedermeyer and F. L. da Silva Electroencephalography: Basic Principles, Clinical Applications, and Related Fields. 2005, Lippincott Williams and Wilkins

[13]. V. Mihajlovi, H. Li, B. Grundlehner, J. Penders and A. C. Schouten "Investigating the impact of force and movements on impedance magnitude and EEG," Proc. IEEE 35th Annu. Int. Conf. Eng. Med. Biol. Soc., 14661469